

Python Basics

What is Python?

Python is a high-level, interpreted, object-oriented, general-purpose programming language developed by Guido van Rossum in 1991.

Why Python?

- Simple and easy to understand
- Wide range of applications
- Dynamically typed programming language
- Supports Object-Oriented Programming (OOP)
- Regular updates and improvements
- Large number of libraries
- Also considered a scripting language

Applications

Web Development, Data Science, Machine Learning, Artificial Intelligence, Game Development, Desktop GUI Applications, Cyber Security, Networking, Automation.

Drawbacks

- Slow execution speed compared to compiled languages
- High memory usage
- Limited support for mobile application development

How Python Works

Python Code (.py) -> Interpreter reads the code -> Converts to Bytecode -> Python Virtual Machine (PVM) -> Output Displayed